

Qatar-2b

R-0751

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250124 · created 2026-07-28T02:57:21+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460698.6211 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.225 +/- 0.065
Transit depth (Rp/Rs)^2	5.07 +/- 2.94 [%]
Orbital Inclination (inc)	89.74 +/- 1.7
Airmass coefficient 1 (a1)	10.9 +/- 6.54
Airmass coefficient 2 (a2)	-1.32 +/- 0.53
Scatter in the residuals of the lightcurve fit is	79.41 %
Best Comparison Star	None
Optimal Aperture	0.76
Optimal Annulus	5.84
Transit Duration (day)	0.036 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.225 $\pm$ 0.065 vs 0.16327 (deviation 0.85 σ)
Duration fit vs published	0.036 d vs 0.0767 d (deviation 12.12 σ)
Mid-time O-C	-1930.1 min
Depth SNR	3.1
Residual scatter	79.41 %

Light curve

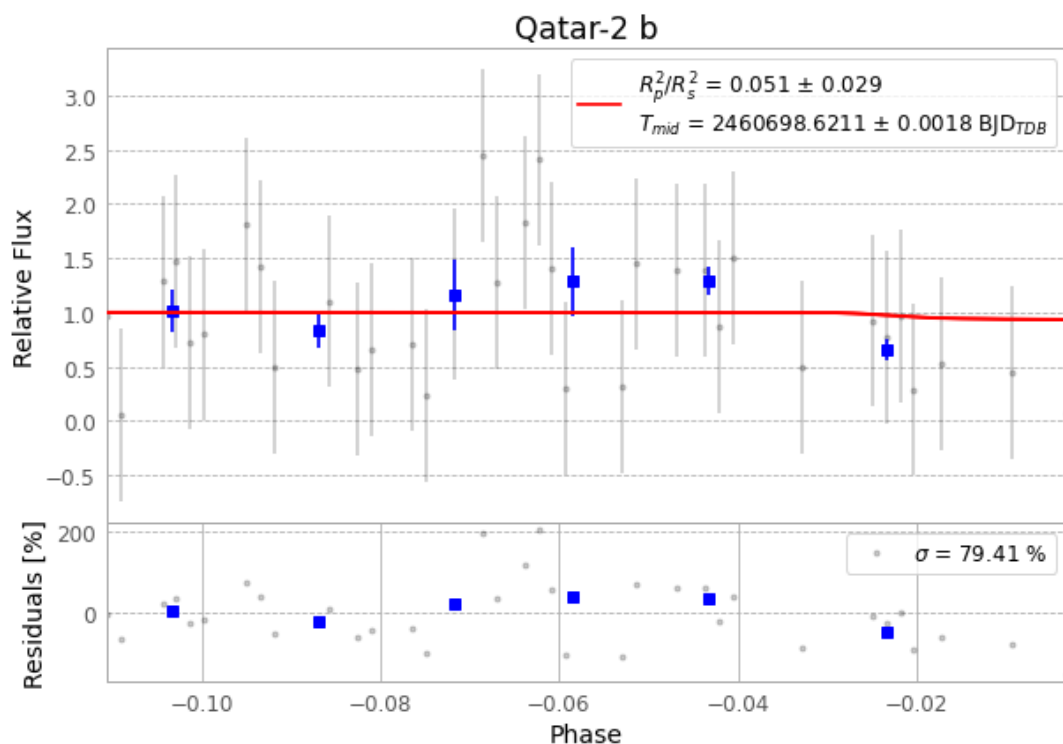


Figure 1 — FinalLightCurve_Qatar-2 b_2025-01-24.png (SHA-256 d09eb81b57d65df0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [496, 315], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d71c56ee0271098...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), Qatar-2250124072116.FITS → Qatar-2250124105416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250124.log (131f0fef10f7...), FinalLightCurve_Qatar-2 b_2025-01-24.png (d09eb81b57d6...), FinalLightCurve_Qatar-2 b_2025-01-24.csv (c35a109cd377...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:57Z.